



Day 2

Synchronous Learning

AI-900 Certification

Preparation

Implementing Partner:



Daftar Materi

1. Get started with generative AI in Azure
2. Introduction to Natural Language Processing Concepts
3. Get started with natural language processing in Azure
4. Get started with speech in Azure

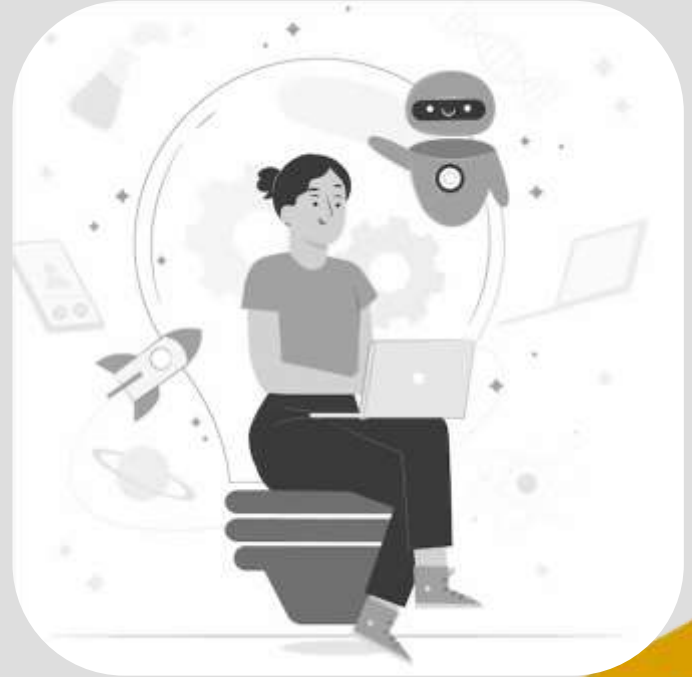


Bab 1

Get started with Generative AI in Azure

Daftar Isi

1. Understanding generative AI applications
2. Understand tools to develop generative AI
3. Understand Azure AI Foundry's model catalog
4. Understand Azure AI Foundry capabilities
5. Understand observability



Bab 1

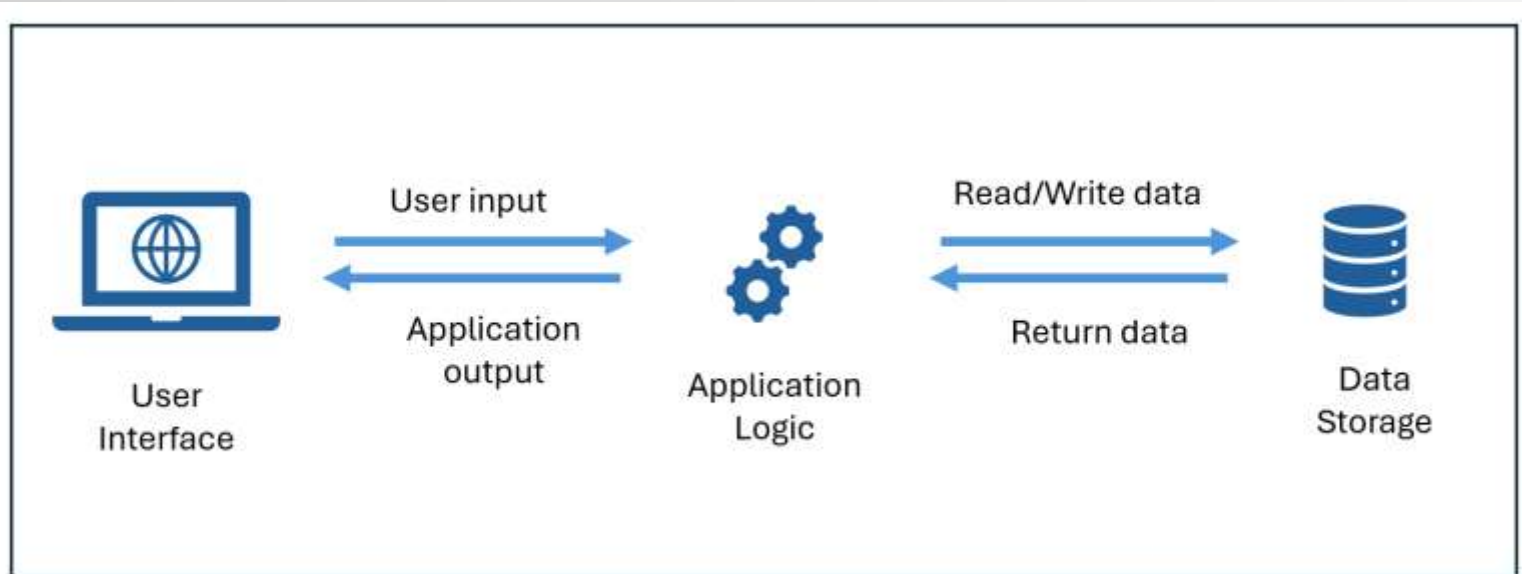
Sub-bab 1

Understand Generative AI Applications



Generative AI Application

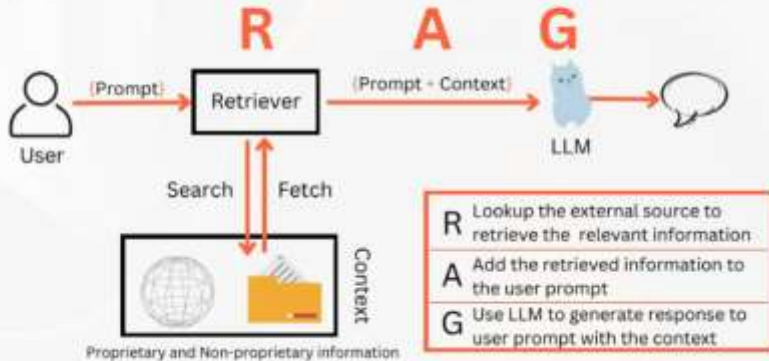
Generative AI applications are built with language models. These language models power the 'app logic' component of the interaction between users and generative AI.



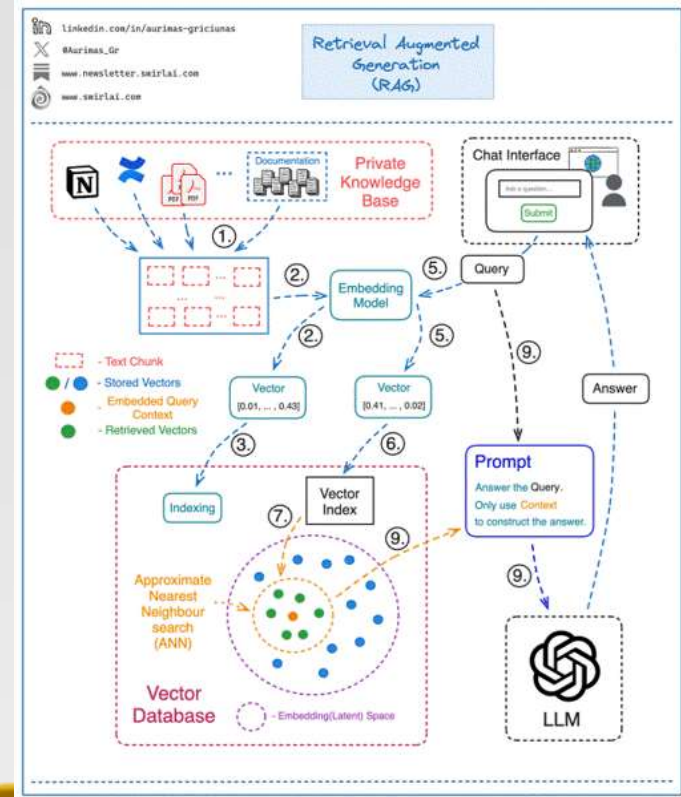
Retrieval Augmented Generation (RAG)

So, What is RAG?

In 2023, RAG has become one of the most used technique in the domain of Large Language Models.



What is RAG?





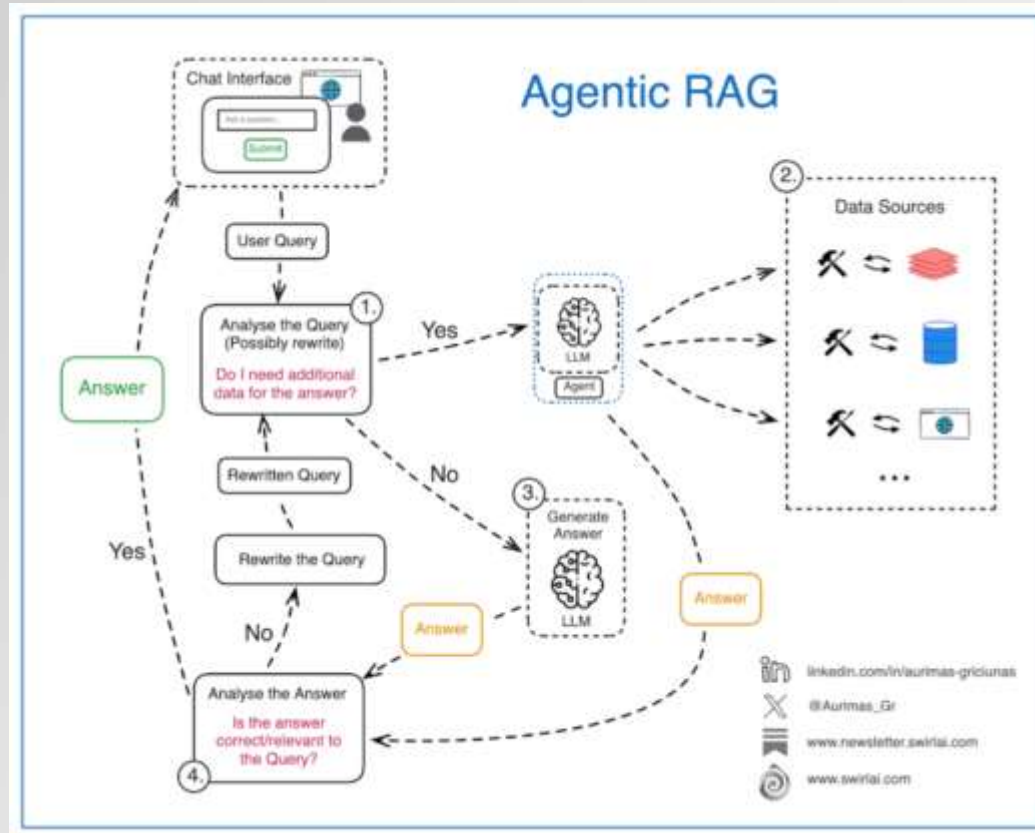
Agents

Agents are applications that can **respond** to user input or **assess situations autonomously**, and **take appropriate actions**. These actions could help with a series of tasks. For example, an "executive assistant" agent could provide details about the location of a meeting on your calendar, then attach a map or automate the booking of a taxi or rideshare service to help you get there.

Agents contain three main components:

- **A language model** that powers reasoning and language understanding
- **Instructions** that define the agent's goals, behavior, and constraints
- **Tools**, or functions, that enable the agent to complete tasks

Agentic RAG



Microsoft AI Solution



Applications

 Microsoft 365

 Microsoft Dynamics 365

Partner Solutions

Application Platform

AI Builder



Power BI



Power Apps



Power Automate



Copilot Studio

Scenario-Based Services



Bot Service



AI Search



Document Intelligence



Video Indexer



Metrics Advisor



Immersive Reader

Customizable AI Models



Vision



Speech



Language



Decision

Azure OpenAI Service

ML Platform



Azure Machine Learning



Category	Description
Ready-to-use	These applications are ready-to-use generative AI applications. They do not require any programming work on the user's end to utilize the tool. You can start simply by asking the assistant a question.
Extendable	Some ready-to-use applications can also be extended using your own data. These customizations enable the assistant to better support specific business processes or tasks. Microsoft Copilot is an example of technology that is ready-to-use and extendable.
Applications you build from the foundation	You can build your own assistants and assistants with agentic capabilities starting from a language model.

Often, you will use services to extend or build generative AI applications. These services provide the infrastructure, tools, and frameworks necessary to develop, train, and deploy generative AI models. For example, Microsoft provides services such as Copilot Studio to extend Microsoft 365 Copilot and Microsoft Azure AI Foundry to build AI from different models.

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Understand Tools to Develop Generative AI





Understand tools to develop generative AI

Microsoft offers a powerful ecosystem of tools and services for building generative AI solutions, designed to support developers, data scientists, and enterprises at every stage of the AI lifecycle. You can develop generative AI solutions with several Microsoft solutions.

This module will focus on Azure AI Foundry, Microsoft's unified platform for enterprise AI operations, model builders, and application development.





Azure AI Foundry's Components

Component	Description
Azure AI Foundry model catalog	A centralized hub for discovering, comparing, and deploying a wide range of models for generative AI development.
Playgrounds	Ready-to-use environments for quickly testing ideas, trying out models, and exploring Azure AI services.
Azure AI services	In Azure AI Foundry portal, you can build, test, see demos, and deploy Azure AI services.
Solutions	You can build agents and customize models in Azure AI Foundry portal.
Observability	Ability to monitor usage and performance of your application's models.

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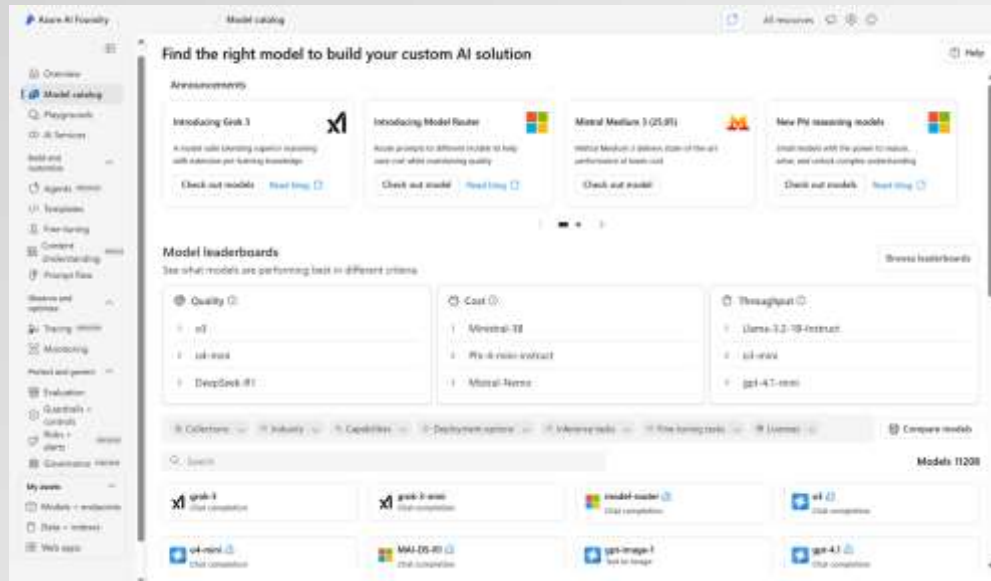
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Understand Azure AI Foundry's Model Catalog



Model Catalog

Azure AI Foundry provides a comprehensive and dynamic marketplace containing models sold directly by Microsoft and models from its partners and community.



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Understand Azure AI Foundry Capabilities





Customizing Models

Method	Description
Using grounding data	Grounding refers to the process of ensuring that a system's outputs are aligned with factual, contextual, or reliable data sources. Grounding can be done in various ways, such as linking the model to a database, using search engines to retrieve real-time information, or incorporating domain-specific knowledge bases. The goal is to anchor the model's responses to these data sources, enhancing the trustworthiness and applicability of the generated content.
Implementing Retrieval-Augmented Generation (RAG)	RAG augments a language model by connecting it to an organization's proprietary database. This technique involves retrieving relevant information from a curated dataset and using it to generate contextually accurate responses. RAG enhances the model's performance by providing it with up-to-date and domain-specific information, which helps in generating more accurate and relevant answers. RAG is useful for applications where real-time access to dynamic data is crucial, such as customer support or knowledge management systems.
Fine-tuning	Involves taking a pretrained model and further training it on a smaller, task-specific dataset to make it more suitable for a particular application. This process allows the model to specialize and perform better at specific tasks that require domain-specific knowledge. Fine-tuning is useful for adapting models to domain-specific requirements, improving accuracy, and reducing the likelihood of generating irrelevant or inaccurate responses.
Managing security and governance controls	Security and governance controls are needed to manage access, authentication, and data usage. These controls help prevent the publication of incorrect or unauthorized information.

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Understand Observability





Observability

There are many ways to measure generative AI's response quality. In general, you can think of three dimensions for evaluating and monitoring generative AI. These include:

- **Performance and quality evaluators:** assess the accuracy, groundedness, and relevance of generated content.
- **Risk and safety evaluators:** assess potential risks associated with AI-generated content to safeguard against content risks. This includes evaluating an AI system's predisposition towards generating harmful or inappropriate content.
- **Custom evaluators:** industry-specific metrics to meet specific needs and goals.



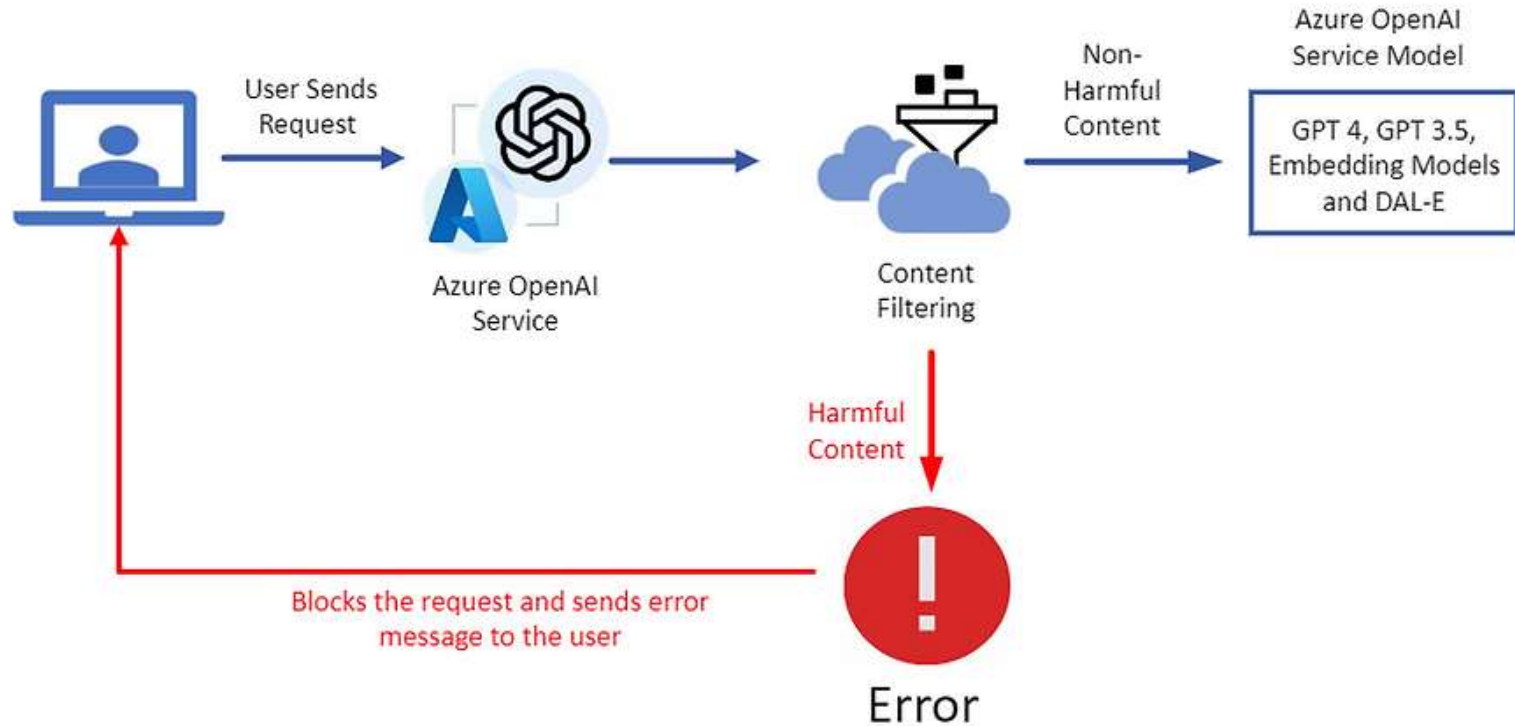
Evaluators

Evaluators are specialized tools in Azure AI Foundry that measure the quality, safety, and reliability of AI responses.

Some evaluators include:

- **Groundedness:** measures how consistent the response is with respect to the retrieved context.
- **Relevance:** measures how relevant the response is with respect to the query.
- **Fluency:** measures natural language quality and readability.
- **Coherence:** measures logical consistency and flow of responses.
- **Content safety:** comprehensive assessment of various safety concerns.

Content Filtering



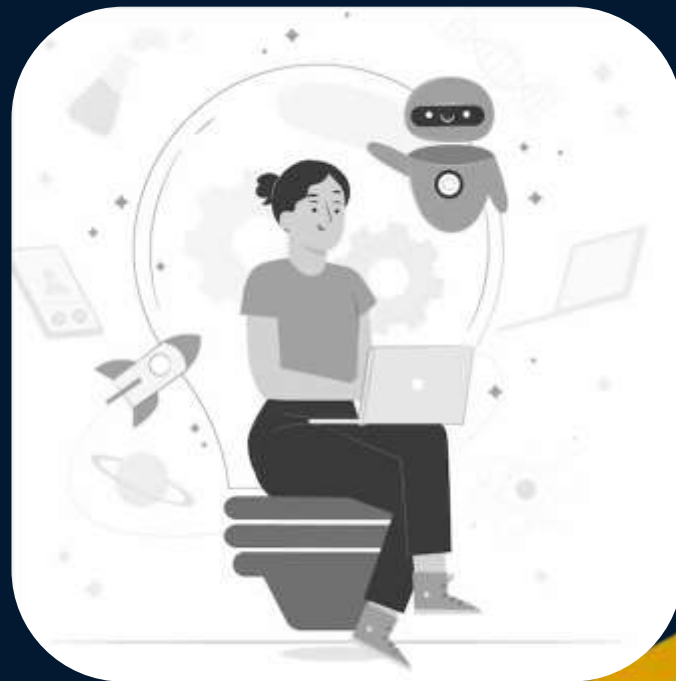
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Bab 2

Introduction to Natural Language Processing Concepts

Daftar Isi

1. Introduction
2. Understand how language is processed
3. Understand statistical techniques for NLP
4. Understand semantic language models



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Introduction



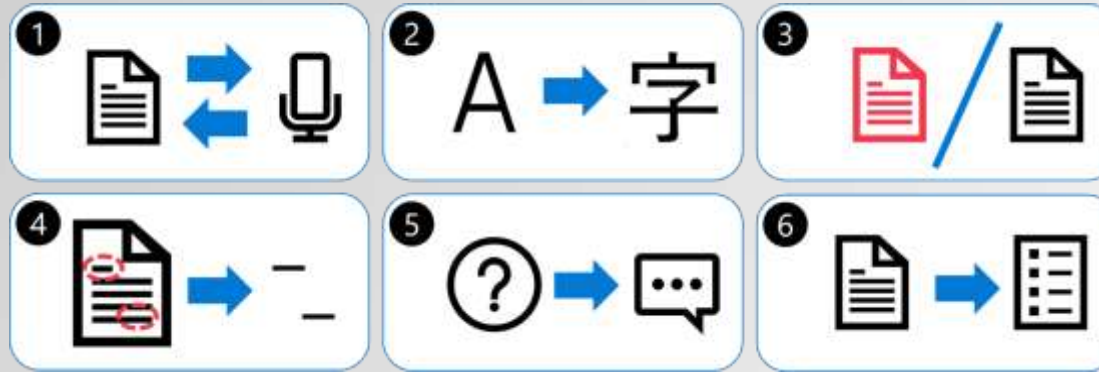


NLP

In order for computer systems to interpret the subject of a text in a similar way humans do, they use natural language processing (NLP), an area within AI that deals with understanding written or spoken language, and responding in kind.

Historically, NLP has been challenging as our language is complex and computers find it hard to understand text. In this module, you learn how developments in AI and specifically NLP have led to the models we use today.

NLP Text Analysis Use Cases



1. **Speech-to-text and text-to-speech** conversion. For example, generate subtitles for videos.
2. **Machine translation**. For example, translate text from English to Japanese.
3. **Text classification**. For example, label an email as spam or not spam.
4. **Entity extraction**. For example, extract keywords or names from a document.
5. **Question answering**. For example, provide answers to questions like "What is the capital of France?"
6. **Text summarization**. For example, generate a short one-paragraph summary from a multi-page document.

Sub-bab 2

Understand How Language is Processed





Corpus

Some of the earliest techniques used to analyze text with computers involve statistical analysis of a **body of text (a corpus)** to infer some kind of semantic meaning. Put simply, if you can determine the most commonly used words in a given document, you can often get a good idea of what the document is about.

In the context of NLP, a **corpus** refers to a large and structured collection of text documents that is used for machine learning tasks. Corpora (plural of corpus) serve as essential resources for training, testing, and evaluating various NLP models.

Corpus



-

Token

—

Sentence

≡

Paragraph



Document



Corpus



Tokenization

The first step in analyzing a corpus is to break it down into tokens. For the sake of simplicity, you can think of each distinct word in the training text as a token, though in reality, tokens can be generated for partial words, or combinations of words and punctuation.

For example, consider this phrase from a famous US presidential speech: "we choose to go to the moon".

Notice that "to" (token number 3) is used twice in the corpus. The phrase "we choose to go to the moon" can be represented by the tokens {1,2,3,4,3,5,6}.



Some Concepts that Affect Tokenization

Concept	Description
Text normalization	Before generating tokens, you may choose to <i>normalize</i> the text by removing punctuation and changing all words to lower case. For analysis that relies purely on word frequency, this approach improves overall performance. However, some semantic meaning may be lost - for example, consider the sentence "Mr Banks has worked in many banks.". You may want your analysis to differentiate between the person "Mr Banks" and the "banks" in which he has worked. You may also want to consider "banks." as a separate token to "banks" because the inclusion of a period provides the information that the word comes at the end of a sentence
Stop word removal	Stop words are words that should be excluded from the analysis. For example, "the", "a", or "it" make text easier for people to read but add little semantic meaning. By excluding these words, a text analysis solution may be better able to identify the important words.
n-grams	Multi-term phrases such as "I have" or "he walked". A single word phrase is a unigram , a two-word phrase is a bi-gram , a three-word phrase is a tri-gram , and so on. By considering words as groups, a machine learning model can make better sense of the text.
Stemming	A technique in which algorithms are applied to consolidate words before counting them, so that words with the same root, like "power", "powered", and "powerful", are interpreted as being the same token.

Sub-bab 3

Understand Statistical Techniques for NLP





Statistical Techniques for NLP

Naïve Bayes is a statistical technique that was first used for email filtering. To learn the **difference between spam and not spam**, two documents are compared. Naïve Bayes **classifiers** identify which tokens are correlated with emails labeled as spam. In other words, the technique finds which group of words only occurs in one type of document and not in the other. The group of words is often referred to as bag-of-words features.

For example, the words *miracle cure*, *lose weight fast*, and *anti-aging* may appear more frequently in spam emails about dubious health products than your regular emails.



Term Frequency – Inverse Document Frequency (TF-IDF)

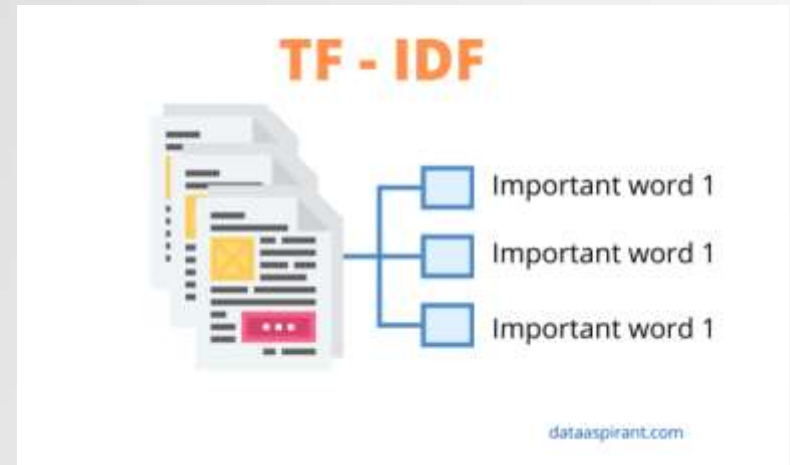
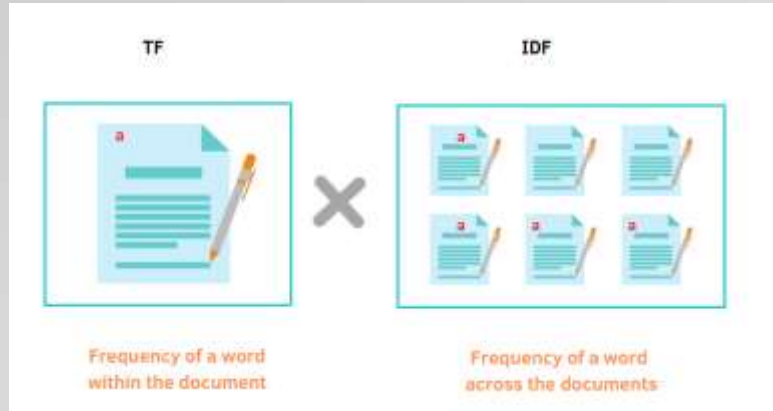
The Term Frequency – Inverse Document Frequency (TF-IDF) technique had a similar approach in that it **compared the frequency of a word in one document with the frequency of the word in a whole corpus of documents.** By understanding in which context a word was being used, documents could be classified based on certain topics. TF-IDF is often used for information retrieval, to help understand which relative words or tokens to search for.



Term Frequency – Inverse Document Frequency (TF-IDF)

TF-IDF calculates scores based on how often a word or term appears in one document compared to its more general frequency across the entire collection of documents. Using this technique, a high degree of relevance is assumed for words that appear frequently in a particular document, but relatively infrequently across a wide range of other documents.

TF-IDF



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Understand Semantic Language Models



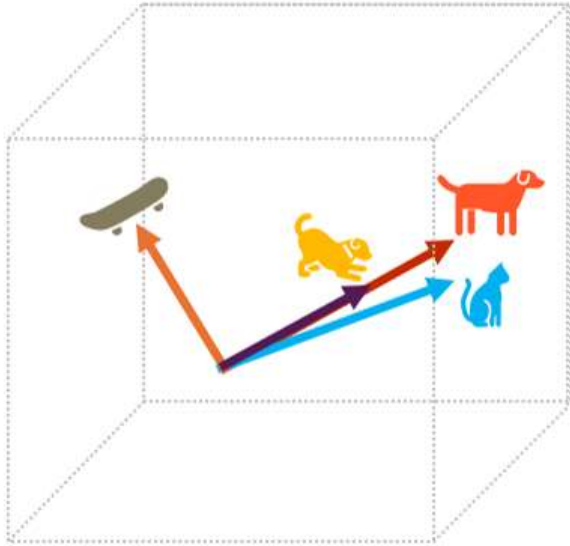


Embeddings

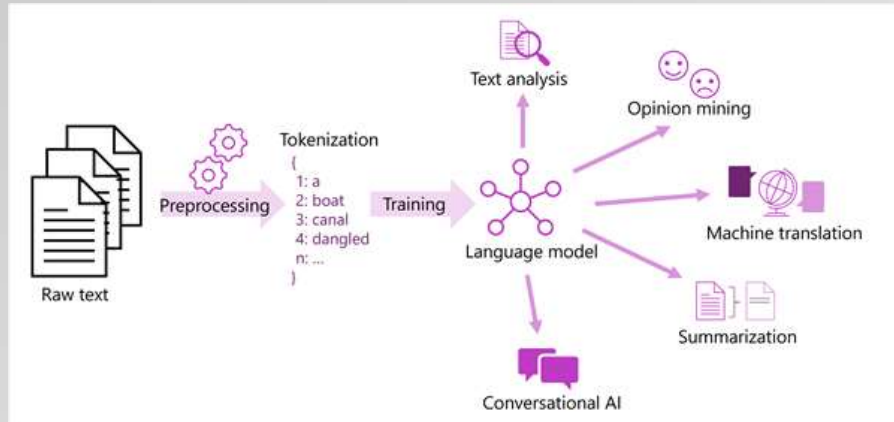
As the state of the art for NLP has advanced, the **ability to train models that encapsulate the semantic relationship between tokens has led to the emergence of powerful deep learning language models.** At the heart of these models is the encoding of language tokens as vectors (multi-valued arrays of numbers) known as embeddings.

Vectors represent lines in multidimensional space, describing direction and distance along multiple axes. Overall, the vector describes the direction and distance of the path from origin to end. Semantically similar tokens should result in vectors that have a similar orientation – in other words they point in the same direction.

A decorative graphic at the bottom right of the slide consisting of several overlapping, wavy, translucent lines in shades of yellow, orange, and red, creating a sense of motion and depth.



	Token	Word	Embedding
	4	dog	[10,3,2]
	8	cat	[10,3,1]
	9	puppy	[5,2,1]
	10	skateboard	[-3, 3, 2]



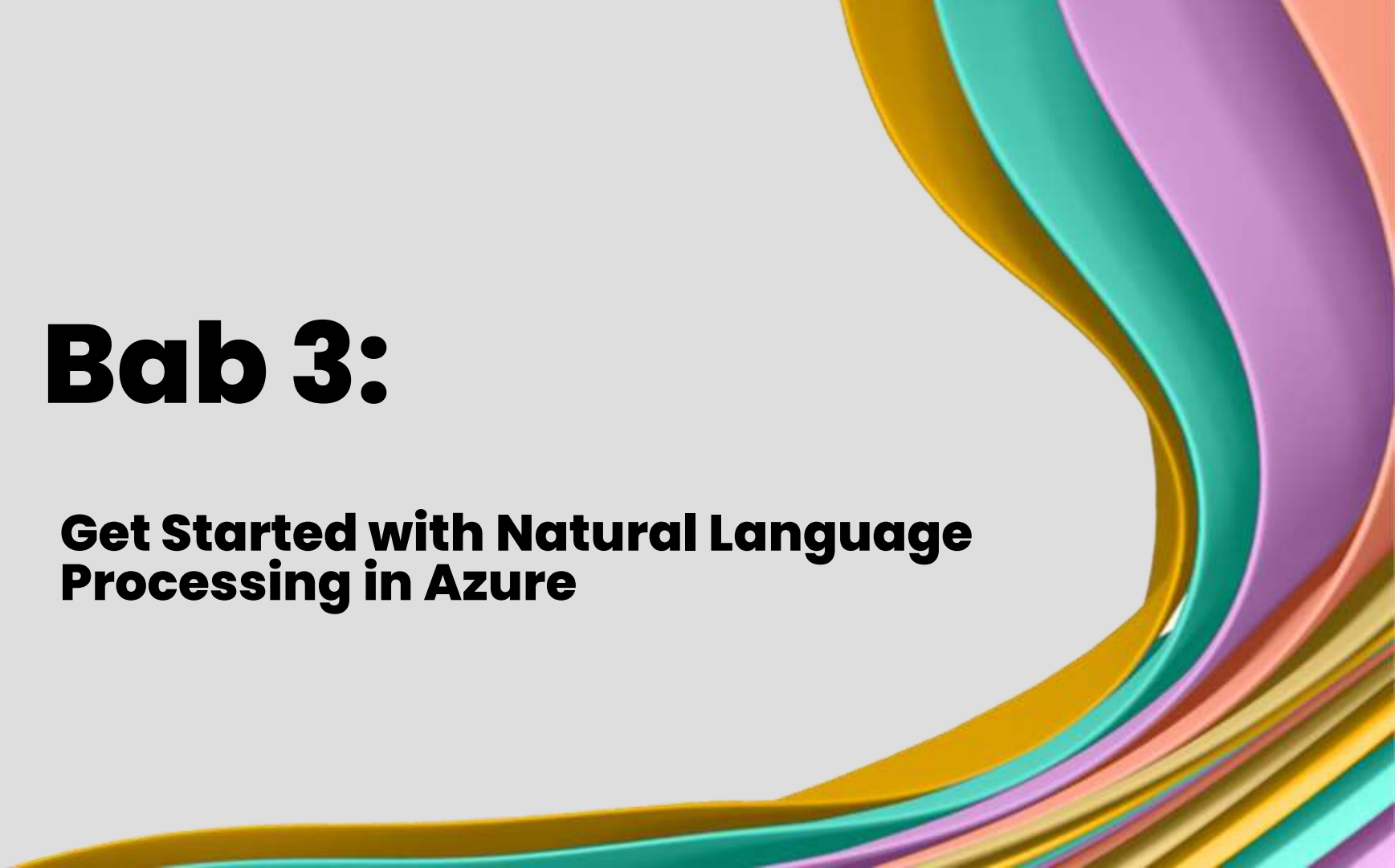
A generalized view of most modern natural language processing solutions is shown in the following diagram. A large corpus of raw text is tokenized and used to train language models, which can support many different types of natural language processing task.

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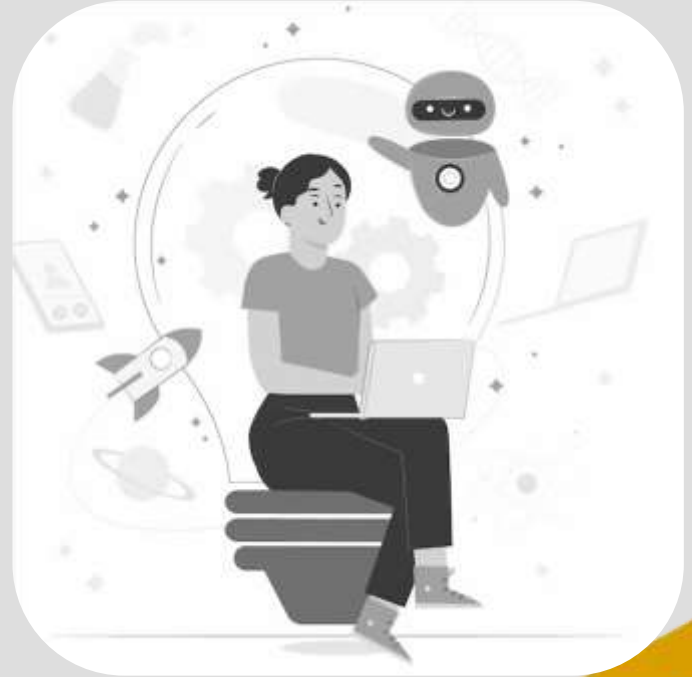
Bab 3:

**Get Started with Natural Language
Processing in Azure**



Daftar Isi

1. Understanding natural language processing in Azure
2. Understand Azure AI Language's text analysis capabilities
3. Azure AI Language's conversational AI capabilities
4. Azure AI Translator capabilities
5. Get Started in Azure AI Foundry



Sub-bab 1

Understand Language Processing on Azure





NLP on Azure



Azure AI Language
Service

A cloud-based service that includes features for understanding and analyzing text. Azure AI Language includes various features that support **sentiment analysis, key phrase identification, text summarization, and conversational language understanding.**



Azure AI Translator
Service

A cloud-based service that uses Neural Machine Translation (NMT) for translation, which analyzes the semantic context of the text and renders a **more accurate and complete translation** as a result.

Sub-bab 2

Understand Azure AI Language's Text Analysis Capabilities





Azure AI Language

Azure AI Language is a part of the Azure AI services offerings that can perform advanced natural language processing over unstructured text. Azure AI Language's text analysis features include:

- **Named entity recognition** identifies people, places, events, and more. This feature can also be customized to extract custom categories.
- **Entity linking** identifies known entities together with a link to Wikipedia.
- **Personal identifying information (PII) detection** identifies personally sensitive information, including personal health information (PHI).
- **Language detection** identifies the language of the text and returns a language code such as "en" for English.
- **Sentiment analysis** and opinion mining identifies whether text is positive or negative.
- **Summarization** summarizes text by identifying the most important information.
- **Key phrase extraction** lists the main concepts from unstructured text.

Sub-bab 3

Azure AI Language's Conversational AI Capabilities





Question Answering

Hi. I'm the Adatum support bot. How can I help you?

Adatum Support at 10:50 AM

I have a question about my bill

You

OK. What's your account number?

Adatum Support at 10:50 AM

123-45-678A

You

Alright. I've found your details.
Is your question about:

1. The bill amount
2. The due date
3. Something else

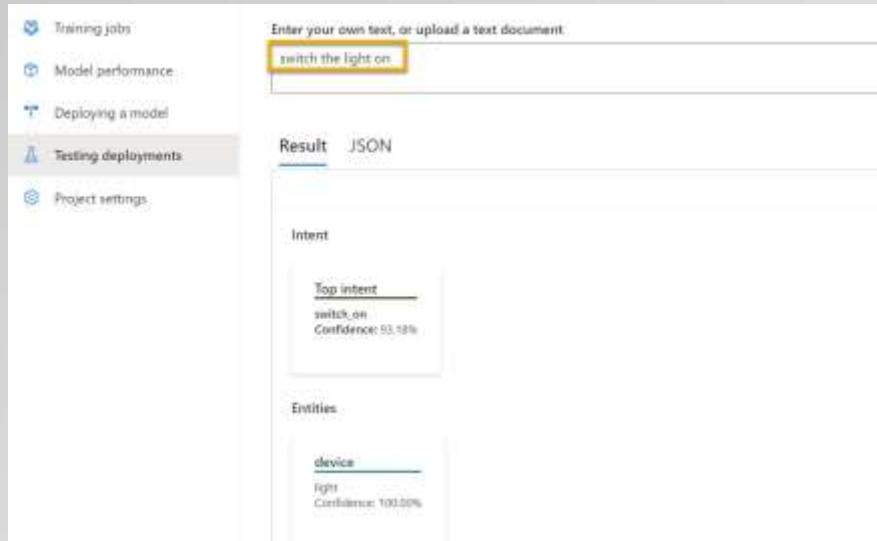
Enter 1, 2, or 3

Type your message here ...

Azure AI Language's question answering feature provides you with the ability to create conversational AI solutions. Question answering supports natural language AI workloads that require an automated conversational element.



Conversational Language Understanding



Azure AI Language service supports conversational language understanding (CLU). You can use CLU to build language models that interpret the meaning of phrases in a conversational setting.

Sub-bab 4

Azure AI Translator Capabilities





Azure AI Translator

Azure AI Translator supports text-to-text translation between more than 130 languages. When using Azure AI Translator, you can specify one from language with multiple to languages, enabling you to simultaneously translate a source document into multiple languages.



Capabilities

Azure AI Translator includes the following capabilities:

- Text translation - used for quick and accurate text translation in real time across all supported languages.
- Document translation - used to translate multiple documents across all supported languages while preserving original document structure.
- Custom translation - used to enable enterprises, app developers, and language service providers to build customized neural machine translation (NMT) systems.

Sub-bab 5

Get Started in Azure AI Foundry





Get Started

Azure AI Language and Azure AI Translator provide the building blocks for incorporating language capabilities into applications. As one of many Azure AI services, you can create solutions in several ways including:

- The Azure AI Foundry portal
- A software development kit (SDK) or REST API



Resources

To use Azure AI Language or Azure AI Translator in an application, you must provision an appropriate resource in your Azure subscription. You can choose either a single-service resource or a multi-service resource.

- **A Language resource** - choose if you only plan to use Azure AI Language services, or if you want to manage access and billing for the resource separately from other services.
- **A Translator resource** - choose if you want to manage access and billing for each service individually.
- **An Azure AI services resource** - choose if you plan to use Azure AI Language in combination with other Azure AI services, and you want to manage access and billing for these services together.

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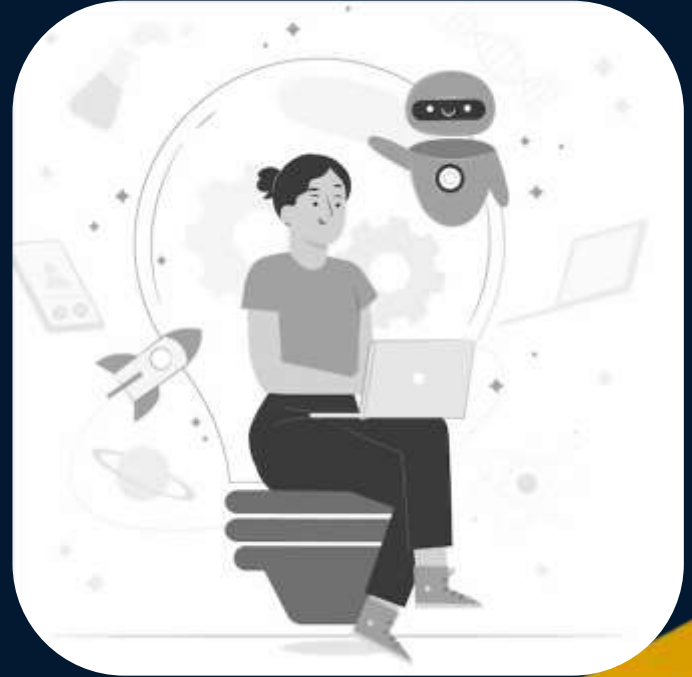


Bab 4

Get Started with Speech in Azure

Daftar Isi

1. Introduction
2. Understand speech recognition and synthesis
3. Get started with Speech on Azure
4. Use Azure AI Speech



Bab 4
Sub-bab 1
Introduction



Introduction

Azure AI Speech provides **speech to text**, **text to speech**, and **speech translation** capabilities through speech recognition and synthesis. You can use prebuilt and custom Speech service models for a variety of tasks, from transcribing audio to text with high accuracy, to identifying speakers in conversations, creating custom voices, and more.

Bab 4

Sub-bab 2

Understand Speech Recognition and Synthesis





Speech Recognition

Speech recognition takes the spoken word and converts it into data that can be processed - often by transcribing it into text. The spoken words can be in the form of a recorded voice in an audio file, or live audio from a microphone. Speech patterns are analyzed in the audio to determine recognizable patterns that are mapped to words. To accomplish this, the software typically uses multiple models, including:

- **An acoustic model** that converts the audio signal into phonemes (representations of specific sounds).
- **A language model** that maps phonemes to words, usually using a statistical algorithm that predicts the most probable sequence of words based on the phonemes.



Speech Synthesis

Speech synthesis is concerned with vocalizing data, usually by converting text to speech. A speech synthesis solution typically requires the following information:

- The text to be spoken
- The voice to be used to vocalize the speech

Bab 4

Sub-bab 3

Get Started with Speech in Azure





Speech to Text

The model that is used by the Speech to text API, is based on the Universal Language Model that was trained by Microsoft. The data for the model is Microsoft-owned and deployed to Microsoft Azure. The model is optimized for two scenarios, conversational and dictation. You can also create and train your own custom models including acoustics, language, and pronunciation if the pre-built models from Microsoft don't provide what you need.

Real-time transcription: Real-time speech to text allows you to transcribe text in audio streams. You can use real-time transcription for presentations, demos, or any other scenario where a person is speaking.

Batch transcription: Not all speech to text scenarios are real time. You might have audio recordings stored on a file share, a remote server, or even on Azure storage. You can point to audio files with a shared access signature (SAS) URI and asynchronously receive transcription results.



Text to Speech

Speech synthesis voices: When you use the text to speech API, you can specify the voice to be used to vocalize the text. This capability offers you the flexibility to personalize your speech synthesis solution and give it a specific character.

The service includes multiple pre-defined voices with support for multiple languages and regional pronunciation, including **neural voices** that leverage neural networks to overcome common limitations in speech synthesis with regard to intonation, resulting in a more natural sounding voice. You can also develop **custom voices** and use them with the text to speech API

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**Akhir Dari
Day 1 Synchronous Learning**

Thank you!

Implementing Partner:

